XIAMEN UNIVERSITY MALAYSIA
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CYS103 Linear Algebra

Chapter 1

Linear Equations and Linear Transformations

- 1.1 System of Linear Equations
- 1.2 Echelon and Reduced Echelon Forms
- 1.3 Solution Sets of Linear Systems
- 1.4 Linear Transformations



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1.3 Solution Sets of Linear Systems

Note: $\vec{x}$, $\underset{\sim}{x}$, $\overrightarrow{x}$

	Definition
Vector (or column vector)	• A matrix with only one column. • Example: $\mathbf{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ in R^2 $\quad \mathbf{v} = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$
R^n (read as: r-n)	• It is the real numbers that appear as entries in the vector. • Example: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ → It is R^3
Trivial solution	• It is the solution where all variables are ZERO. • Example: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$ → $x_1 = 0, x_2 = 0$, and $x_3 = 0$
Nontrivial solution	• It is any solution where at least one variable $(x_1, x_2, \ldots)$ is NOT zero.
Basic variable	• A variable that corresponds to a pivot column. (a column with a leading 1 in REF)
Free variable	• A variable that does NOT correspond to a pivot column.

$\vec{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ or $\vec{x} = (-1, 2)$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0$

→ free variable
basic variable

1.3 Solution Sets of Linear Systems

Example (trivial solution, basic variable):

Determine whether the following system has only the trivial solution.

$$\begin{aligned} x_1 + x_2 &= 0 \\ x_1 - x_2 &= 0 \end{aligned}$$

Solution: ↓ augmented matrix

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\vdots$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Therefore:

$$x_1 = 0$$
$$x_2 = 0$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0$$

∴ The system has only the trivial sol. ∎

1.3 Solution Sets of Linear Systems

Example (nontrivial solution, basic variable, free variable):

Determine whether the following system has the nontrivial solution.

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 0 \\ 2x_1 + 4x_2 + 2x_2 &= 0 \end{aligned}$$

Solution: augmented matrix:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Then, $x_1 + 2x_2 + x_3 = 0 \longrightarrow x_1 = -2x_2 - x_3$.

$0 = 0$

x_2 is a free variable.

x_3 is a free variable

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ It has the nontrivial solutions.

not trivial

basic & free variable.

infinitely many solutions

when $x_2 = 0$, $x_3 = 0$.

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0$$

basic variable

trivial col.

$\{(0,0,0), (\ ), (\ ), \ldots\}$

1.3 Solution Sets of Linear Systems

Extra

Example (nontrivial solution, basic variable, free variable):

Determine whether the following system has the nontrivial solution.

$$\begin{aligned} x_1 &+ 2x_2 + x_3 = 0 \\ 2x_1 &+ 4x_2 + 2x_2 = 0 \end{aligned}$$

$A\vec{x} = 0$

homogeneous linear system

Solution:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

1.3 Solution Sets of Linear Systems

❑ Homogeneous Linear Systems

mxn matrix A :

a vector : $\vec{x}$

basic V. + free V.

Definition

The **homogenous equation** $A\mathbf{x} = 0$ has a **nontrivial solution** if and only if the equation has **at least one free variable**, where A is an $m \times n$ matrix, $\mathbf{x}$ is an n column vector of variables, and 0 is the zero vector in R^m.

Example:

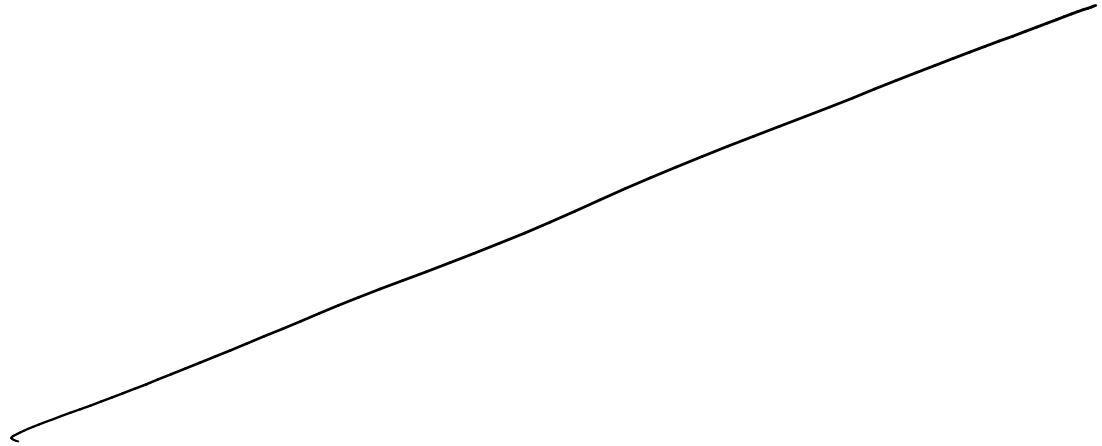
$$\begin{aligned} x_1 + 2x_2 + x_3 &= 0 \\ 2x_1 + 4x_2 + 2x_2 &= 0 \end{aligned}$$

$$A \quad \vec{x} = 0$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

#.

- A **linear system** is **homogenous** if it can be expressed in the homogenous equation form $A\mathbf{x} = 0$.
- This system $A\mathbf{x} = 0$ always **has at least one solution**, that is, $\mathbf{x} = 0$ (the zero vector in R^n).
  - ➢ This **zero solution** is called the **trivial solution**.



1.3 Solution Sets of Linear Systems

Example:

$$\begin{aligned} x_1 + 10x_2 &= 0 \\ 2x_1 + 20x_2 &= 0 \end{aligned}$$

can be written as:

1.3 Solution Sets of Linear Systems

Example:

Determine the following homogeneous system has a nontrivial solution.

$$3x_1 + 5x_2 - 4x_3 = 0$$
$$-3x_1 - 2x_2 + 4x_3 = 0$$
$$6x_1 + x_2 - 8x_3 = 0$$

augmented matrix.

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ The system has the nontrivial solution.

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

x_3 : $-4/3$ column

x_3 : pivot -

$$\begin{cases} x_1 - \frac{4}{3}x_3 = 0 \\ x_2 = 0 \\ 0 = 0 \end{cases} \quad \text{basic. U} \quad \begin{cases} x_1 = \frac{4}{3}x_3 \\ x_2 = 0 \\ x_3 \text{ is a free variable.} \end{cases}$$

when $x_3 = 0$.

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix} \quad \#.$$

$\{(0,0,0), (4/3,0,1), \cdots\}$ —— solution set.

($x_3 = 1$)

1.4 Linear Transformations

❑ The Matrix Equation $A\mathbf{x} = \mathbf{b}$

a vector : [ ]

Definition

If A is an $m \times n$ matrix, which columns $\mathbf{a_1}, \mathbf{a_2}, \ldots, \mathbf{a_n}$, and if $\mathbf{x}$ is in R^n, then the product of A and $\mathbf{x}$, denoted by $A\mathbf{x}$, is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$,

$$A\mathbf{x} = [\mathbf{a_1} \quad \mathbf{a_2} \quad \cdots \quad \mathbf{a_n}] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a_1} + x_2\mathbf{a_2} + \cdots + x_n\mathbf{a_n} = \mathbf{b}$$

Therefore, $A\mathbf{x} = \mathbf{b}$ is called a **matrix equation**.

Example:

$A \vec{x} = \vec{b}$

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ -5x_2 + 3x_3 &= 1 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

#

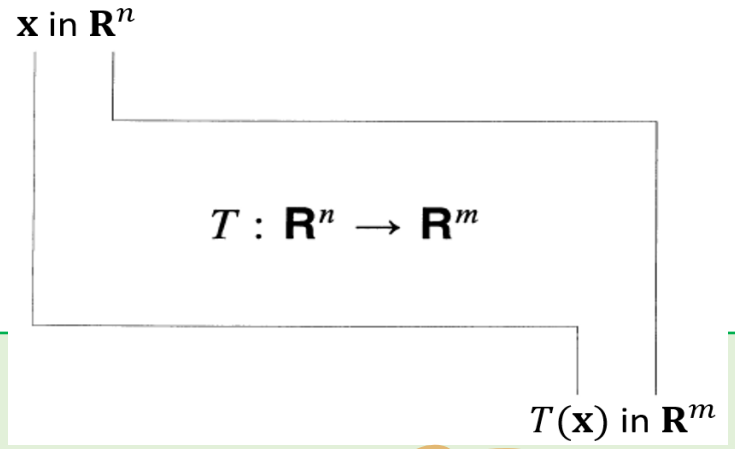
1.4 Linear Transformations

❑ Linear Transformations

$T(\vec{x}) = A(\vec{x})$

Definition

A **transformation** (or function, or mapping) T from R^n to R^m, denoted as $T: R^n \to R^m$ is a rule that assigns to each vector $\mathbf{x}$ in R^n a vector $T(\mathbf{x})$ in R^m.



Note:

- The set R^n is called the **domain** of T.
- R^m is the **codomain** of T.
- For $\mathbf{x}$ in R^n, the vector $T(\mathbf{x})$ in R^m is the **image** of $\mathbf{x}$.
- The set of all images $T(\mathbf{x})$ is called the **range** of T.

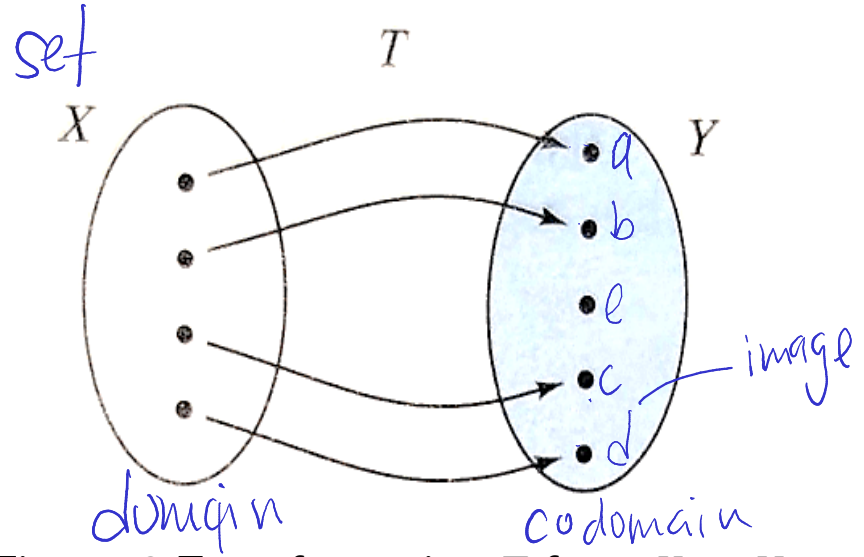


Figure 2 Transformation T from X to Y.

1.4 Linear Transformations

Example:

Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T: R^2 \to R^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

a) Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T.

$$T(\bar{u}) = A(\bar{u}) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} (1\times2)+(-3\times(-1)) \\ (3\times2)+(5\times(-1)) \\ (-1\times2)+(-1\times7) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

R^2 R^3

b) Find an $\mathbf{x}$ in R^2 whose image under T is $\mathbf{b}$.

$T(\bar{x}) = \bar{b} \longrightarrow A(\bar{x}) = \bar{b}$

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \; ?$$

$A\vec{x} = \vec{b}$

augmented matrix.

$$\begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \end{array} \begin{bmatrix} x_1 & x_2 & c \\ 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix}$$

ERO

$$\begin{bmatrix} x_1 & x_2 & \\ 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

Linear equation 1 : R_1

2 : R_2

3 : R_3

Therefore, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$ ※.

1.4 Linear Transformations

A **transformation T is linear** if it has the properties:

Properties

A **transformation T** is linear if

i) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in domain of T.

ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T.

Note: Every matrix transformation is a linear transformation.

Definition

A **transformation T from R^n to R^m** is called a **linear transformation** if there exists an $m \times n$ matrix A such that

$$T(\mathbf{x}) = A\mathbf{x}$$

for all $\mathbf{x}$ in the vector space R^n.

1.4 Linear Transformations

Example:

Define a linear transformation $T: R^2 \to R^2$ by $T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$. Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

$$T(\bar{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \checkmark$$

$$T(\bar{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

$$T(\bar{u}+\bar{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

$\neq$ OR

$$T(\bar{u}+\bar{v}) = T(\bar{u}) + T(\bar{v})$$

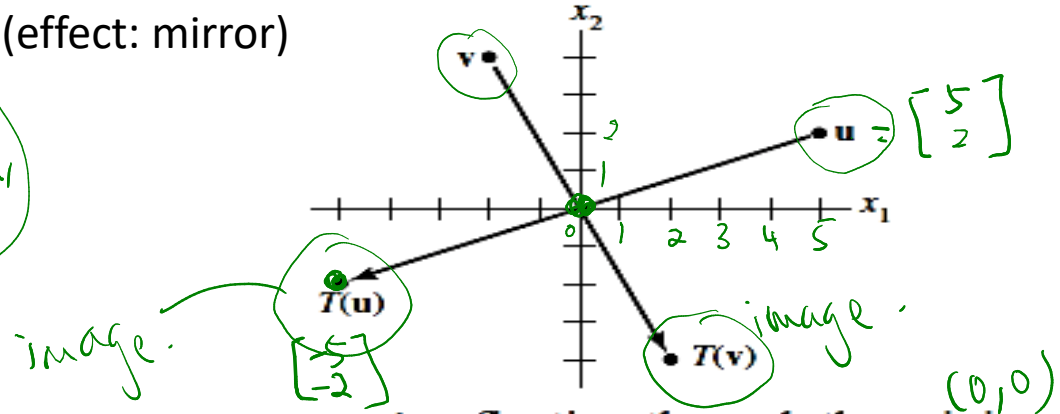
$$\begin{bmatrix} -4 \\ 6 \end{bmatrix} = ?$$

$$\begin{bmatrix} -4 \\ 6 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} + \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

1.4 Linear Transformations

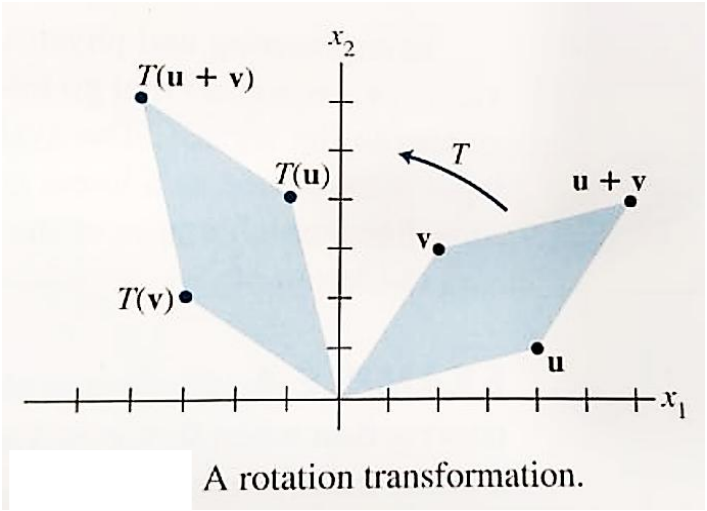
❑ The Matrix of A Linear Transformations

Reflection
(effect: mirror)



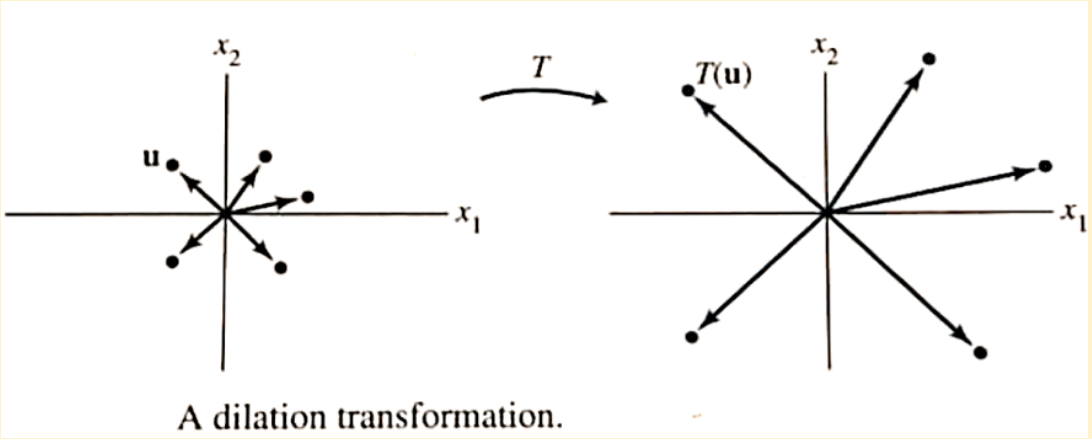
A reflection through the origin.

Rotation
(effect: turn)



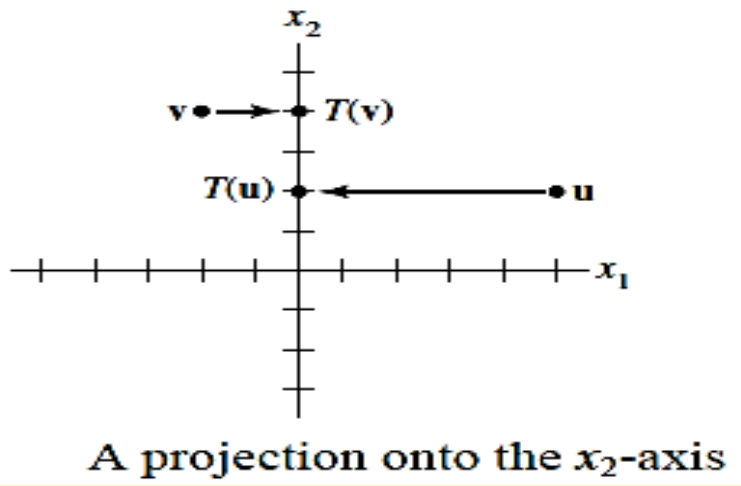
A rotation transformation.

Dilation (effect: zoom in/out)



A dilation transformation.

Projection
(effect: flatten)



A projection onto the x_2-axis

1.4 Linear Transformations

- Transformation T is **one-to-one** if and only if $T(\mathbf{x}) = 0$ has the **trivial solution** ($\mathbf{x} = 0$).

Example:

Let $T(x_1, x_2) = (x_1 - 2x_2, 4x_1, 3x_1 + 2x_2)$. Show that $T(\vec{x}) = A(\vec{x})$ and T is one-to-one linear transformation.

① $T(\bar{x}) = \begin{matrix} R1 \\ R2 \\ R3 \end{matrix}\begin{bmatrix} 1 & -2 \\ 4 & 0 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A(\bar{x})$ (x_1, x_2; R_1, R_2, R_3)

$\therefore \quad T(\bar{x}) = A(\bar{x})$

② $\begin{bmatrix} 1 & -2 \\ 4 & 0 \\ 3 & 2 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ (pivot; x_1, x_2; 10, 20, 0; e.g. $x_1 = 10$, $x_2 = 20$)

Therefore, T is one-to-one linear transformation. #.

(x_1 and x_2 are basic variables)

The End.